

Charley Sanchez

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Research Summary

I work at the intersection of machine learning, security, and healthcare. My current research at UMBC studies the robustness of multimodal foundation models for physiological signals such as EEG, ECG, and wearable data, focusing on how they respond to realistic sensor faults and attacks. Previously, I worked on causal treatment-effect estimation, wearable-based time-series modeling for mental health assessment, and privacy-aware embedded vision. I am broadly motivated by difficult and interesting problems in the clinical AI space, combining statistical, deep learning, and systems methodologies to find solutions that hold up in practice.

Education

University of Maryland, Baltimore County 2026–Present
Doctor of Philosophy in Computer Science

UCLA Samueli School of Engineering 2025
Master of Engineering in Artificial Intelligence

- **GPA:** 3.74/4.00
- **Relevant Coursework:** Machine Learning, Deep Learning, Linear Programming

University of California, Santa Barbara 2023
Bachelor of Science in Physics

- **Relevant Coursework:** Linear Algebra, Differential Equations, Probability and Statistics, Computational Physics

Research Experience

Graduate Researcher — UMBC, Prof. Dong Li Lab (FSI Lab) 2026–Present
Robustness of Multimodal Physiological Foundation Models

- Building a threat-model-driven evaluation of multimodal foundation models for biosignals (EEG, ECG, EOG, EMG) proposed for clinical and wearable health monitoring.
- Defining realistic attacks and sensor faults at the level of the physical signal, including ones that target a single input stream rather than the whole model.
- Applying attacks consistently across model architectures and measuring their impact with statistically careful, subject-level evaluation to identify which design decisions make multimodal health models fragile or resilient.

Graduate Researcher — Georgia Institute of Technology, Prof. Anind K. Dey Lab 2026
AI for Mental Health

- Developed modeling pipelines for mental health assessment using wearable and behavioral time-series data; completed project and delivered codebase and documentation to the incoming research cohort.
- Investigated image-based representations of time series, including Markov Transition Fields (MTF) and Gramian Angular Fields (GAF), to enable convolutional and multimodal learning approaches.
- Designed a feature selection pipeline using gradient boosted trees to identify clinically significant predictors within the raw database prior to MTF/GAF encoding, targeting more focused CNN training under limited data volume constraints.
- Analyzed how representation choices affect downstream classification stability, interpretability, and sensitivity to temporal irregularities in clinical signals.

Research Volunteer — Johns Hopkins University, Prof. Wenbo Wu Lab 2025–2026
Causal Inference and Double Machine Learning

- Studied how the choice of deep nuisance architectures (DCN, ResMLP, Transformers) in partially linear logistic DML impacts bias, variance, and stability of treatment-effect estimates.
- Implemented cross-fitting, orthogonalized scores, and multi-stage nuisance modeling in PyTorch to estimate multi-treatment effects under finite-sample constraints.
- Developed reproducible SLURM workflows to run 500–2000+ model configurations across seeds and hyperparameter grids with structured logging for downstream analysis.
- Showed that optimized gamma-model design improved recovery of true treatment effects by 10–22% in nonlinear synthetic settings.

EHR Representation Learning for Healthcare Cost Modeling

- Initiated a structured literature review on cost-supervised embedding methods for medical diagnosis codes (ICD/CPT) in EHR data, targeting improved patient cost estimation for Medicare Advantage plans.
- Began investigating how learned diagnosis representations can improve upon standard actuarial baselines for risk adjustment and plan cost modeling.

Graduate Researcher — UCLA, Prof. Nader Sehatbakhsh Lab

2025

Privacy-Aware Vehicular Autonomous Computation (PAVAC)

- Developed and analyzed a real-time face anonymization pipeline on Jetson Orin Nano, focusing on the accuracy/latency/privacy trade-offs of different detection and anonymization strategies.
- Designed mosaic and noise-based anonymization that reduced compute by over 92% while preventing face reconstruction and preserving navigation-relevant visual signal.
- Developed an evaluation pipeline using SAM-derived masks (Dice 0.827, Recall 0.897) with ~41ms end-to-end latency under deployment-like conditions.
- Integrated the anonymization module into a physical rover stack and validated robustness on CARLA simulation edge cases; co-author on manuscript in preparation.

Publications & Manuscripts

H. Khalili, P. Do, A. Dabiran, **C. Sanchez**, A. Hafemeister, V. Nguyen, K. Apicharttrisorn, and N. Sehatbakhsh. *Argus: Real-Time Privacy-Preserving Video Streaming for Delivery Robots*. Manuscript in preparation.

Awards

UCLA MEng Fellowship Award

2024

Awarded based on academic performance, merit, and financial need as part of the Master's program.

Professional Experience

Contracted Software Consultant — Econ One Research

2026

Privacy-Aware Local LLM Infrastructure

- Designed and deployed a self-hosted LLM system for a legal-sector economic consulting firm, enabling privacy-preserving inference for expert witness workflows in antitrust litigation without data exposure to external APIs.
- Implemented a vLLM, Open WebUI, and LiteLLM stack with architecture designed for extensibility, including planned integration of Continue.dev for agentic coding workflows.
- Advised on model selection, access control, and deployment tradeoffs to meet client-specific confidentiality requirements in legal proceedings.

Software Engineer — Econ One Research

2024

- Designed a PostgreSQL-based document system with automated metadata indexing, reducing query latency by ~40% for large discovery datasets.

- Built a multithreaded document processing pipeline (ThreadPoolExecutor) that increased ingestion throughput by 3x while preserving auditability.
- Developed internal tools for time-tracking and billing workflows that improved reporting accuracy to over 95% and reduced manual reconciliation effort.

Lab Assistant — Pacific Diagnostic Laboratories

2023–2024

- Accessioned and processed 500+ specimens per day across chemistry, hematology, and microbiology benches while maintaining specimen integrity and turnaround targets.
- Used the EPIC EHR and laboratory information systems to track specimen status and verify results, working with nurses and clinicians to resolve missing or delayed orders.
- Followed HIPAA and institutional protocols when handling sensitive patient data and authored updated bench SOPs that reduced new technician onboarding time by ~30%.

Selected Projects

Generalization in sEMG Keystroke Recognition

github.com/SOTA-4

- Implemented a bidirectional LSTM with residual connections for sEMG-based keystroke recognition on the EMG2QWERTY dataset (346 hours, 108 users).
- Achieved a 12% CER improvement over baseline by applying domain transfer and adaptation techniques across users and recording sessions.

Skills

Methods: Adversarial robustness evaluation, causal machine learning (DML, orthogonalization), EHR representation learning, biosignal and time-series modeling, sequence models (LSTMs, Transformers), real-time inference

ML Systems: PyTorch, ONNX Runtime, TensorRT, FastAPI, Docker, Linux, SLURM

Clinical Systems: EPIC EHR, laboratory information systems, HIPAA-compliant data handling

Programming: Python, C/C++ , SQL, Bash

LLMs/RAG: vLLM, Open WebUI, LiteLLM, LangChain, embedding-based retrieval, AWS Bedrock (boto3, IAM)